

Pentium Microprocessor

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Lecture References:

- ▶ Book:

- ▶ *Microprocessors and Interfacing: Programming and Hardware*, Chapter # 15, **Author:** Douglas V. Hall
- ▶ *The Intel Microprocessor 8086...Arch. Prog, Interfacing.* **Author:** Bary, Bray
- ▶ *Microprocessor and Microcomputer – Based System Design*, **Author:** Mohamed Rafiquzzaman

- ▶ Lecture Materials:

- ▶ *M.A. Sattar, BUET, Dhaka, Bangladesh.*

Introduction of Pentium

- ▶ The concepts of 80386 microprocessor and 80387 coprocessor together evolved the 80486 microprocessor.
 - ▶ **Only new idea here is the introduction of 8K cache**
 - ▶ The cache is used for storing both *data* and *instructions*
- ▶ Pentium was introduced in 1993
- ▶ Pentium processor is an improvement to the architecture found in 80486.

Pentium Improvements

- ▶ Improved cache structure
 - ▶ Reorganized to form **two level (L2) caches** that are each **8K bytes in size**
 - ▶ One for caching data
 - ▶ Another for caching instructions ★
- ▶ Wider data bus width
 - ▶ Increased from **32 bit to 64 bits**
- ▶ Faster numeric processor
 - ▶ Operates about **5 times faster** than the 80486 numeric processor

Pipelining in Pentium

- ▶ Pentium has two pipelines
- ▶ **U pipeline**
 - ▶ U-pipeline can execute any Pentium instruction
- ▶ **V pipeline**
 - ▶ V-pipeline only executes only *simple instructions*
- ✓ ▶ Each pipeline has 5 stages
 - ▶ i. Pre-fetch
 - ▶ ii. Instruction Decode
 - ▶ iii. Address Generation
 - ▶ iv. Execute, Cache, and ALU Access
 - ▶ v. Write back

Pipelining Stages in Pentium

▶ **Pre-fetch:**

- ▶ Instructions are fetched from the Instruction cache and aligned in prefetch buffers for decoding

▶ **Instruction Decode:**

- ▶ Instructions are decoded into the Pentium's internal instruction format.

▶ **Address Generation:**

- ▶ Address computations take place at this stage

▶ **Execute, Cache, and ALU Access:**

- ▶ The integer hardware executes the instruction

▶ **Write-back:**

- ▶ The results of the computation are written back to the register

Super Scalar Machine

- ▶ Any Processor capable of parallel instruction execution of multiple instructions is known as **superscalar machine**.
- ▶ As, in Pentium, there are **two** execution lines --- U-line and V-line, so Pentium is a ***Super Scalar Processor***
- ▶ What are the differences? to understand the concepts in terms of processor execution stages:
 - ▶ Parallelism
 - ▶ Simultaneity – Simultaneously Parallel
 - ▶ Pipelining – Parallel and Simultaneous

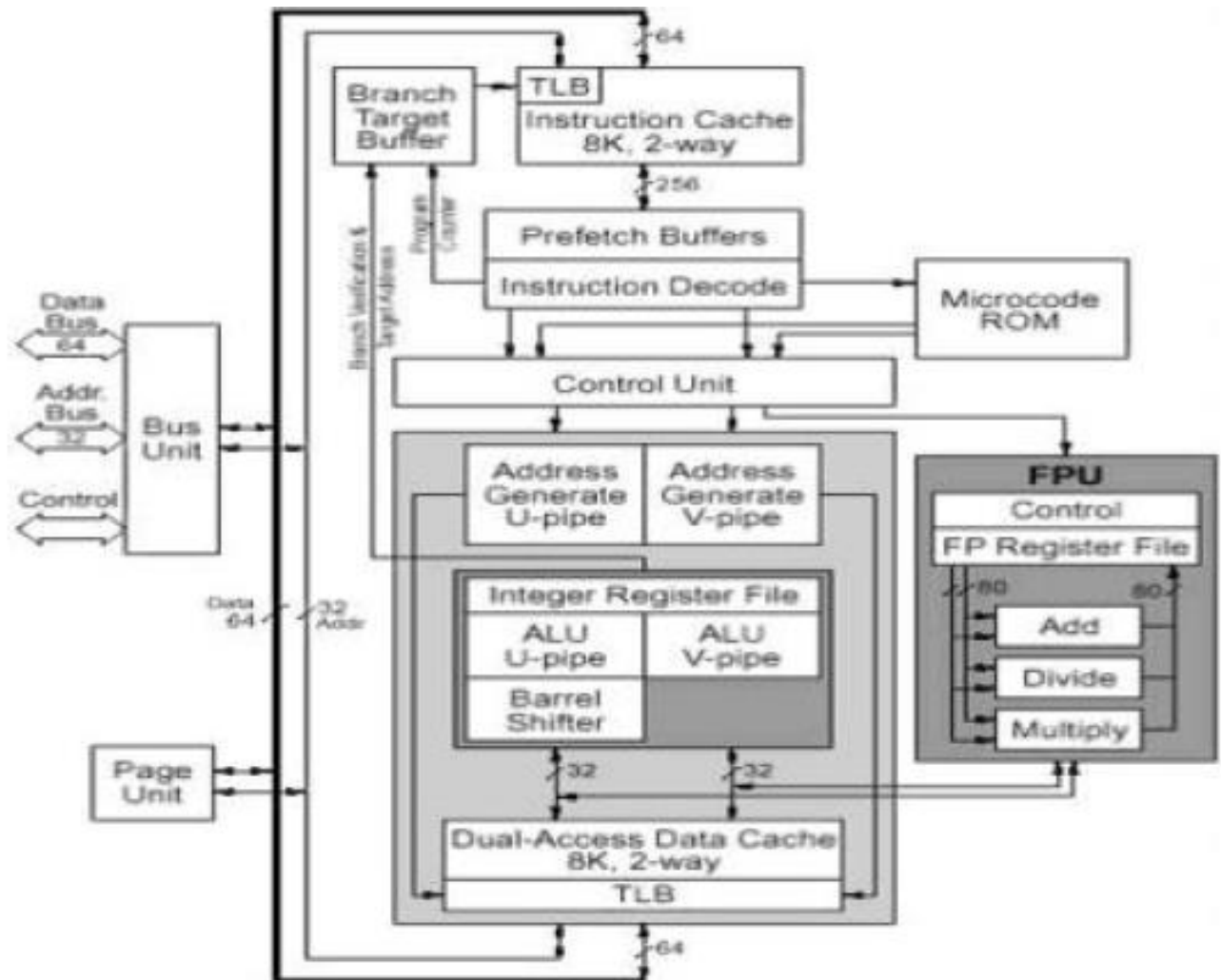
Pentium Architecture

▶ It has all the units similar to 80386.

- ▶ *Instruction unit*
- ▶ **Segmentation unit**
- ▶ **Paging unit**
- ▶ *Bus unit*
- ▶ *Execution unit*

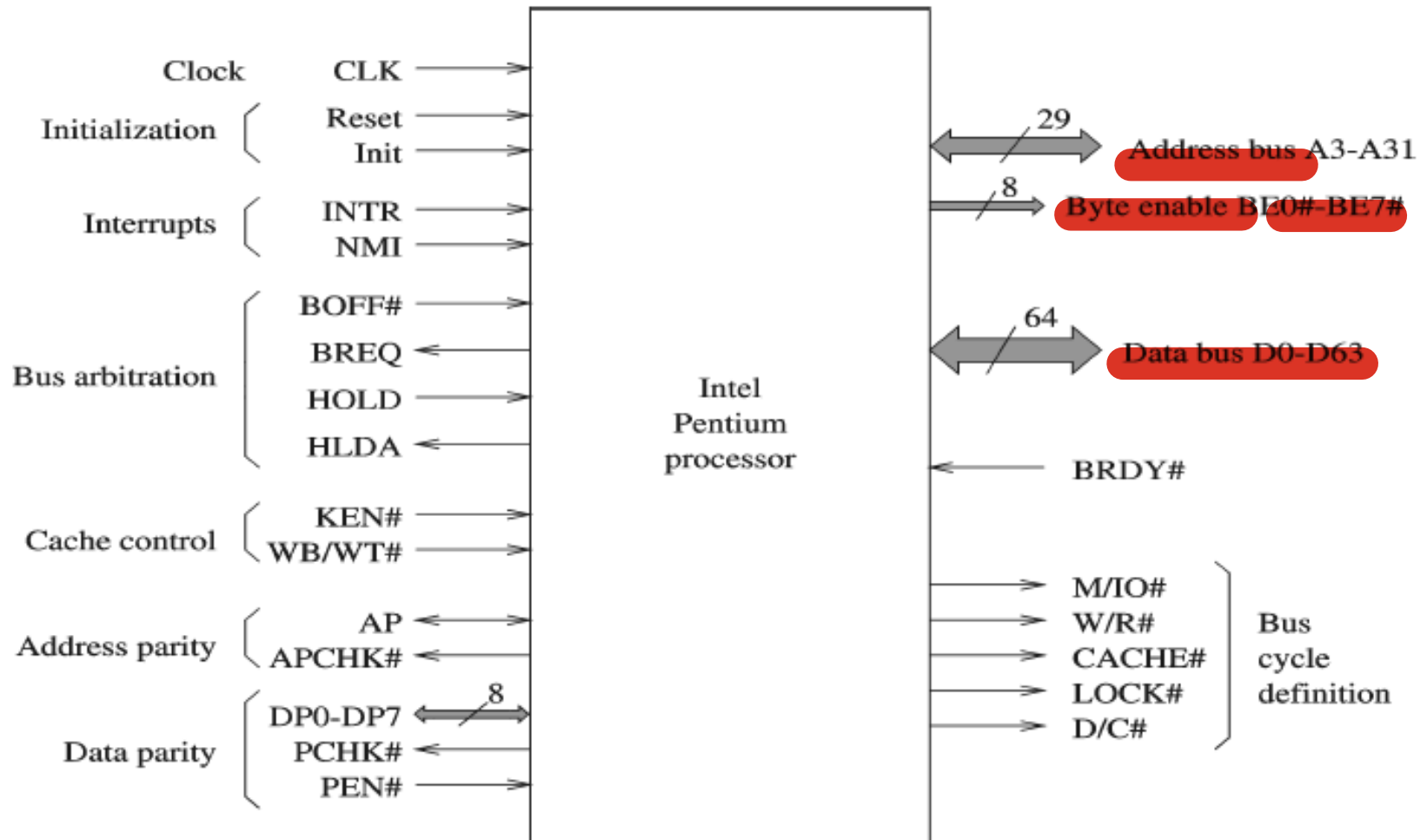
▶ The newer one introduced as-

- ▶ *Floating Point unit (FPU)*



Pentium Architecture

► Pin Diagram



Pentium Architecture:

Floating Point Unit (FPU)

- ▶ FPU has 8 80-bit general purpose floating point registers, ST(0) through ST(7)
- ▶ **It has 8-stage pipeline**
 - ▶ First 5 stages are identical to U and V pipelines
 - ▶ 2 additional execution stages
 - ▶ First execution stage (X1 stage)
 - ▶ Second execution stage (X2 stage)
 - ▶ In these two stages FPU reads the data from data cache and executes the floating point computation
 - ▶ One additional error reporting stage

Pentium Architecture: **Pins**

- ▶ Packaged in 273 pins
 - ▶ Data Bus 64 pins
 - ▶ Address Bus 29 pins plus 3 pins
 - ▶ Control Bus 75 pins
 - ▶ Vcc + Ground 99 pins
 - ▶ No Connection (NC) 6 pins

Pentium Data Bus

- ▶ 64-bit data bus
- ▶ D0 through D63 (Data Bus)
- ▶ Bi-Directional
- ▶ These signals make up the Pentium's 64-bit bidirectional data bus.

Pentium Address Bus

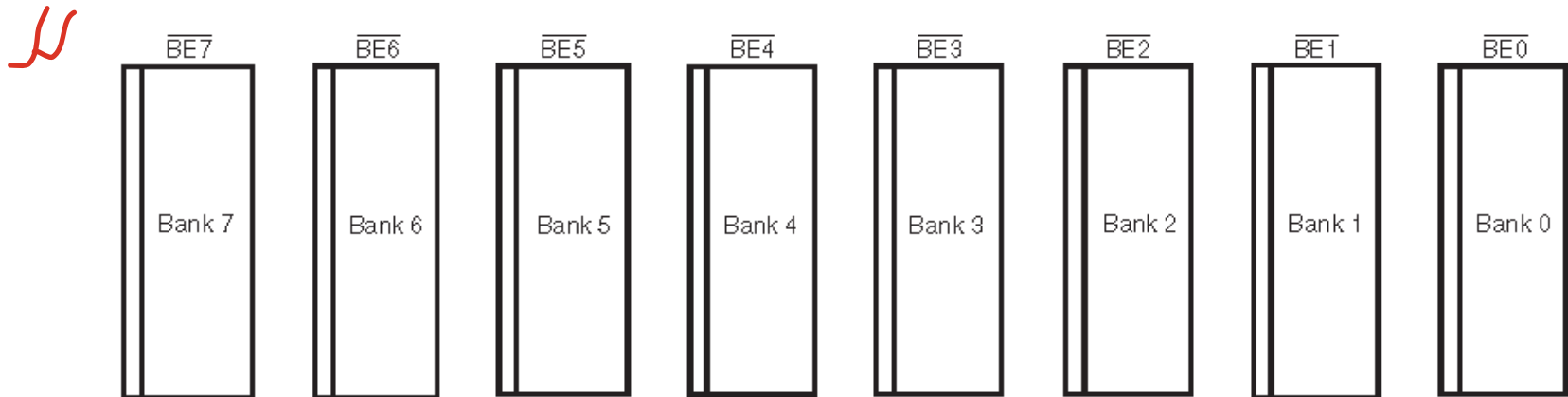
- ▶ 32-bit address bus.
- ▶ A3 through A31 (Address Lines)
 - ▶ Output/Input --- bi-directional
- ▶ These 29 address lines, together with the byte enable outputs *BE0-BE7*, form the Pentium's 32-bit address bus.
- ▶ A memory space of 4 GB is possible, along with 65536 I/O ports.
- ▶ The address lines are used as input during an inquire cycle to read an address *into the Pentium*, for examination by the internal cache.

Pentium Address Bus

- ▶ No A0, A1 and A2 address lines are available in Pentium
 - ▶ They are internally decoded to produce BE0 to BE7 signals
- ▶ **BE0 through BE7**
 - ▶ Byte (Bank???) Enable lines
 - ▶ Memory are arranged in total 8 banks
- ▶ **Output**
 - ▶ These, together with A3 through A31, make up the 32-bit address output by the Pentium.
 - ▶ Each byte enable is used to control a different 8-bit portion of the processor's ~~64-bit data bus~~.
 - ▶ BE0 enables Bank 0

Pentium Memory System

- ▶ Pentium's memory arranged in 8 banks
- ▶ Each bank stores **byte-wide data with a parity bit**
 - ▶ **It helps for error detection and correction in data**
- ▶ BE0 to BE7 selects the banks
- ▶ New feature added to Pentium is its capability to generate and check parity for address bus



Features of All Pentium Processors



Pentium Pro: Salient Features

- ▶ The notable difference in the Pentium Pro than the earlier Pentium is that
 - ▶ There are provisions for a **36-bit address bus**, which allows access to **64G bytes of memory**.
 - ▶ ***This is meant for future use*** because no system today contains anywhere near that amount of memory.
- ▶ Pentium Pro is available in two versions.
 - ▶ One version contains a **256K level 2 cache**;
 - ▶ The other contains a **512K level 2 cache**
- ▶ Pentium Pro microprocessor is packaged in an immense 387-pin PGA (pin grid array).

Pentium II: Salient Features

- ▶ Extension to Pro architecture with some differences
 - ▶ Internal cache in PII has been moved out of the chip
 - ▶ PII is not available as a single chip
 - ▶ Rather is available on a small plug-in circuit board, known as **Cartridge**, along with level 2 (L2) cache chip
- ▶ Various versions are available
 - ▶ **Celeron** is a version without L2 cache
 - ▶ **Xeon** is enhanced by having up to 2M L2 cache

Pentium III: Salient Features

- ▶ Based on Pro architecture, not on Pentium II
- ▶ Like PII, PIII is packaged in a cartridge instead of IC chip
- ▶ Additionally a **Coppermine** is packaged in an IC with 370 pins
- ▶ **Coppermine** is an internal cache with 256K advanced transfer mechanism within the IC running at processor speed

✗ Why not used 512K Cache?

- ▶ *It has been observed that, increasing cache size from 256K to 512K improves the performance by only a few percent*

Pentium III: Salient Features

- ▶ Various versions of Pentium III are also available like Pentium II
 - ▶ Standard Pentium III
 - ▶ Celeron Pentium III uses 66MHz bus speed
 - ▶ Xeon Pentium III allows larger cache for server applications
 - ▶ Still Xeon is popular for server processors

Pentium IV: Salient Features

- ▶ Based on Pro architecture, not on P II or P III
- ▶ P IV is packaged in 421 pins IC
- ▶ It uses physically smaller transistors
 - ▶ Makes it much smaller and faster than P III
- ▶ Released initially in November 2000 with 1.3GHz speed
 - ▶ Was available with speed more than 3GHz

Thank You !!

